

Decision Table Testing Cheat Sheet

Test combinations of conditions systematically, with no gaps.

Introduction

Decision Table Testing is a black-box technique for systems where the outcome depends on combinations of conditions — business rules, eligibility checks, pricing logic. It guarantees you cover every meaningful combination and exposes rules that were never specified. This cheat sheet shows the structure, a worked table and the exam traps.

Anatomy of a Decision Table

Part	Meaning
Conditions	The inputs or circumstances (rows in the top half)
Actions	The outcomes or system responses (rows in the bottom half)
Rules	Each column — a unique combination of condition values and its actions
T / F	Whether a condition is true or false for that rule
X	Marks which action fires for that rule

How Many Rules?

A full (or "complete") decision table has one rule for every combination of conditions. For N binary (true/false) conditions there are 2^N rules. Two conditions give 4 rules, three conditions give 8, and so on. You then collapse rules that produce the same action to simplify.

Worked Example — online discount eligibility

Rule: a customer gets a 10% discount only if they are a registered member AND their basket is over 50 pounds. If they are a member but the basket is 50 or under, they get free delivery only. Non-members get nothing extra.

Conditions	Rule 1	Rule 2	Rule 3	Rule 4
Registered member?	T	T	F	F
Basket over 50?	T	F	T	F

Conditions	Rule 1	Rule 2	Rule 3	Rule 4
Action: 10% discount	X	-	-	-
Action: Free delivery	-	X	-	-
Action: No offer	-	-	X	X

Each column is one test case. Rules 3 and 4 both give "No offer" regardless of basket value, so they could be collapsed into a single rule with a dash ("-") for the basket condition to indicate it does not matter.

Step-by-Step Method

1. List all the conditions that affect the outcome.
2. List all the possible actions.
3. Build a column for every combination of condition values (2^N for binary conditions).
4. Fill in which action(s) each rule triggers.
5. Collapse redundant rules, then derive one test case per remaining rule.

Exam Traps & Common Mistakes

- Miscounting rules — for 3 binary conditions it is 8 rules (2^3), not 6.
- Forgetting non-binary conditions multiply the combinations (2 x 3 conditions give 6 rules).
- Collapsing rules that look similar but actually fire different actions.
- Confusing conditions (inputs) with actions (outputs) when laying out the table.

COVERAGE TIP

Minimum coverage for decision table testing is one test case per rule (column). Collapsing rules reduces the count, but every distinct combination of relevant conditions must still be exercised at least once.

Best Practices

- Use decision tables whenever a requirement reads "if... and... then...".
- Collapse impossible or redundant combinations to keep the table readable.
- Review the table with a business analyst — gaps often reveal missing requirements.

INSIDE STLC PRO TIP

Decision tables are a testing and a requirements tool. Building the full table frequently surfaces combinations the specification never mentioned — raise those as questions before you write a single test.